

MVP Extensions: Contextual Thompson Sampling

William Dennis

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1 Motivation

The MVP agents (Thompson Sampling, ϵ -greedy, UCB, random) are bandits—they select actions based on reward history alone, ignoring the current state. This means the agent learns “nudge is good on average” but cannot learn “nudge is effective when the user is sedentary but pointless when they are already active.”

Contextual Thompson Sampling addresses this by conditioning action selection on state features. The agent maintains separate posterior distributions for each context-action pair, enabling it to learn context-dependent intervention effectiveness.

2 Formulation

2.1 Context

The context feature is the current activity level $c_t \in \{\text{sedentary}, \text{active}\}$, already present in `StateView`. This gives $|\mathcal{C}| = 2$ contexts.

2.2 Posterior Structure

Instead of a single (α_a, β_a) pair per action, the contextual agent maintains:

$$(\alpha_{c,a}, \beta_{c,a}) \quad \forall c \in \mathcal{C}, a \in \mathcal{A}$$

With 2 contexts and 2 actions, this yields 4 posterior pairs (vs. 2 for standard TS).

2.3 Action Selection

Given state s_t with activity $c_t = s_t.\text{activity}$:

$$a^* = \arg \max_{a \in \mathcal{A}} \theta_{c_t,a}, \quad \theta_{c_t,a} \sim \text{Beta}(\alpha_{c_t,a}, \beta_{c_t,a})$$

2.4 Posterior Update

After observing reward r_t :

$$\alpha_{c_t,a_t} \leftarrow \alpha_{c_t,a_t} + \mathbb{I}[r_t > 0], \quad \beta_{c_t,a_t} \leftarrow \beta_{c_t,a_t} + \mathbb{I}[r_t = 0]$$

Only the posterior for the observed (c_t, a_t) pair is updated. Other context-action pairs are unaffected.

2.5 Config

agents:

- type: thompson_sampling
 alpha_prior: 1
 beta_prior: 1
 contextual: true
 context_feature: activity

When `contextual` is omitted or `false`, behaviour is identical to standard Thompson Sampling. Backward compatible.

3 Results

All agents run over 10 random seeds (1–10), 450 timesteps each (90 days $\times$ 5 steps/day). Results are mean $\pm$ standard deviation across seeds.

Agent	Total Reward	Mean Reward/Step
Thompson Sampling (standard)	157.8 $\pm$ 14.8	0.351 $\pm$ 0.033
Thompson Sampling (contextual)	170.8 $\pm$ 15.6	0.380 $\pm$ 0.035
Improvement	+13.0	+8.2%

Table 1: Contextual vs. standard Thompson Sampling on `config/rule_based.yaml` (10 seeds, 450 timesteps). Contextual TS improves by 8.2% by learning that nudges are more effective for sedentary users.

The improvement is modest but consistent. The contextual agent discovers that nudging sedentary users has higher expected reward than nudging active users—a distinction the standard bandit cannot represent.

4 Implementation

4.1 Files Modified

- `src/rl_health_interventions/agents/thompson_sampling.py` – Extended `ThompsonSamplingAgent` with `contextual` and `context_feature` parameters
- `src/rl_health_interventions/config/schemas.py` – Added optional `contextual` and `context_feature` fields to `AgentConfig` with validation
- `tests/unit/agents/test_thompson_sampling.py` – 4 new tests: schema acceptance, regression, context-dependent learning, uniform reward convergence

4.2 Validation

- 165 tests pass (1 skipped – existing data loader skip)
- `ruff check clean`

- `ruff format --check clean`
- Backward compatible: existing configs without `contextual` field produce identical results

5 Design Decisions

Question	Decision	Status
Which context feature?	<code>activity</code> (sedentary/active) – already in <code>StateView</code> , no state changes needed	Confirmed
Why not <code>step_of_day</code> ?	5 discrete values $\times$ 2 actions = 10 posteriors. Manageable but less interpretable than <code>activity</code> . Can be added later.	Deferred
Why not continuous features?	Requires widened <code>StateView</code> + feature pipeline (Phase 2). Binary context is the minimal viable extension.	Deferred
Should the ABC widen to vector in/out?	Not yet. Current string-in/string-out ABC works for discrete contexts. Vector ABC is a Phase 2 concern.	Deferred

Table 2: Design decisions for contextual Thompson Sampling.

6 Next Steps

- **Additional context features:** `step_of_day` (5 values) would give 10 posterior pairs. Requires only a config change.
- **Continuous context:** Requires widened `StateView` (heart rate, step count as floats), widened `TransitionModel` ABC (vector in/out), and feature pipeline. This is Phase 2 work.
- **Learned transitions:** The contextual bandit learns which actions work in which contexts. The next step is learning the transition function itself from data—a separate project requiring real intervention response data (HeartSteps V2, All of Us).